A **Calcium-Looping Direct Air Capture (DAC) system** is an engineered carbon dioxide removal (CDR) technology designed to extract CO_{2} directly from ambient air. It relies on a continuous, reversible chemical cycle using calcium-based minerals (primarily limestone and lime) as a sponge to trap and release carbon dioxide. [1, 2, 3]

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1. The Core Chemical Loop

The entire system operates on three main chemical steps that form a closed loop: [1, 2]

Step 1: Hydration (Activation)

Before the system can efficiently trap CO_{2} from the air, calcium oxide (CaO, or quicklime) is mixed with water ($H_{2}O$) to form calcium hydroxide ($Ca(OH)_2$, or slaked lime). This significantly increases the material's chemical reactivity.

$$\mathrm{CaO + H_2O \rightarrow Ca(OH)_2 + Heat}$$

Step 2: Carbonation (Capture)

The $Ca(OH)_2$ is exposed to ambient air. It reacts naturally with the ultra-dilute CO_{2} in the atmosphere ($\approx 420\text{ ppm}$) to form solid calcium carbonate ($CaCO_{3}$, or limestone) and release the water.

$$\mathrm{Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O}$$

Step 3: Calcination (Release & Regeneration)

The solid $CaCO_{3}$ is collected and fed into a high-temperature kiln called a calciner. Heated to approximately $900^{\circ }\text{C}$, the limestone decomposes, releasing a **100% pure stream of CO_{2}** gas and leaving behind solid CaO.

$$\mathrm{CaCO_3 + Heat \rightarrow CaO + CO_2} \uparrow$$

The regenerated CaO is sent back to **Step 1** to repeat the process, while the pure CO_{2} is compressed and sent for permanent underground geological storage or industrial utilization. [1]

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2. Physical Process Configurations

In practice, companies (such as Heirloom Carbon Technologies) and researchers deploy this chemistry in two main ways: [1]

- **Passive Solid Looping:** The solid calcium powder is spread across large, stacked trays inside modular structures called air contactors. It relies on natural wind currents to bring CO_{2} into contact with the powder over a few days before it is gathered for heating. [1, 2, 3, 4]
- **Liquid-Solid Co-Looping:** Air is passed through a liquid chemical solution (like potassium hydroxide) to quickly grab the CO_{2}. Calcium hydroxide is then introduced to precipitate the captured carbon out as solid $CaCO_{3}$ pellets, which are then calcined. [1, 3]

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3. Key Advantages

- **Abundant and Cheap Sorbent:** The process utilizes limestone ($CaCO_{3}$), which is one of the most naturally abundant, non-toxic, and inexpensive minerals on Earth. [1]
- **Industrial Synergy:** The degraded or spent calcium minerals can be directly diverted into cement manufacturing as raw feedstock. This creates a highly sustainable, net-negative carbon footprint framework for industrial concrete. [1]
- **High Purity Output:** The high-temperature breakdown yields an unpolluted CO_{2} stream that is exceptionally easy to compress, transport, and permanently sequester. [1,

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Instead of pursuing maximum energy efficiency with expensive components, Rivan's core strategy focuses on mass manufacturing and engineering simplicity. They construct their modular production systems inside shipping containers using cheap, commonly available materials.

Rivan Industries utilizes a solid-looping cycle, producing porous calcium hydroxide structures through slaking, extrusion, and quick drying to maximize surface area for CO2 absorption. These engineered, porous blocks are transported on continuous conveyor systems through alternating absorption chambers and high-temperature calcination kilns to create a continuous DAC cycle. Learn more about their technology in

To create a porous calcium hydroxide ($Ca(OH)_2$) structure without collapsing the essential porosity needed for the Direct Air Capture (DAC) reaction, Rivan Industries combines mechanical mixing with materials science engineering.

Your concern is absolutely correct: **if the material is overly compacted, the pores will collapse (low porosity), preventing air from penetrating through. This causes the CO_2 to react only with a thin outer surface layer, leaving the inner core completely "dead" (unreacted).**

The block-shaping process and the solution to the porosity issue are executed as follows:

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1. How the Porous $Ca(OH)_2$ Blocks Are Formed

Instead of using massive hydraulic force (like pressing concrete bricks), Rivan utilizes a **Low-Pressure Extrusion** technique combined with pore-forming additives:

- **Step 1: Hydration and Slaking:** Quicklime (CaO) powder is mixed with water (H_2O) at a precisely calculated ratio to convert it into $Ca(OH)_2$. This highly exothermic reaction naturally generates tiny air bubbles within the molecular matrix as excess water rapidly evaporates.
- **Step 2: Blending Pore-Forming Agents:** To guarantee that the structure does not become a solid, dense mass, a small amount of volatile or combustible organic matter (such as fine wood flour, starch, or foaming surfactants) is mixed directly into the calcium paste.
- **Step 3: Low-Pressure Extrusion:** The paste is pushed through an extruder under minimal pressure to shape it into engineered geometries—typically honeycomb channels or corrugated sheets—maximizing the surface area exposed to the airflow.
- **Step 4: Controlled Drying:** The extruded blocks are dried to evaporate the remaining water and remove the pore-formers. This leaves behind an intricate, sponge-like network of microscopic **micro-pores**.

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2. The Danger of Over-Compaction (Low Porosity)

The consequence of excessive compaction in Calcium Looping technology is known as the **core-plugging phenomenon due to porosity reduction**:

[Over-compacted Block] ➔ Airflow passes ➔ CO2 reacts ONLY with the outer CaO crust ➔ Forms a dense, hard CaCO3 layer ➔ CO2 CANNOT penetrate deeper into the core ➔ Carbon capture efficiency drops by >80%.

- **Molar Volume Increase:** When $Ca(OH)_2$ reacts with CO_2 to transform into calcium carbonate ($CaCO_3$), the volume of the solid molecule **increases by roughly 10% to 20%**.
- **Choking the Air Pathways:** If the block is packed too tightly at the beginning, the superficial molecules will expand as soon as they absorb a small amount of CO_2. This expansion instantly seals off all adjacent micro-pores. Consequently, the outer shell turns into hard limestone, completely isolating the unreacted $Ca(OH)_2$ core inside.

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3. Rivan's Design Solution: Balancing Porosity vs. Structural Durability

An ideal DAC sorbent block must solve a conflicting engineering paradox: **it must be porous enough for air to flow through, yet durable enough not to crumble into dust while moving through the continuous mechanical loop.** Rivan optimizes this balance by:

1. **Macro-Honeycomb Geometries:** The blocks feature large, visible macroscopic channels that let air glide through freely without causing a pressure drop, while the walls of these channels are made of micro-porous structures to trap the CO_2.
2. **Embracing Short Sorbent Lifespans:** Instead of trying to make the calcium blocks indestructible for thousands of cycles (which would require making them too dense and lowering porosity), Rivan accepts that the blocks will gradually degrade over multiple heating-cooling loops. Since limestone is an ultra-cheap mineral, the degraded fine powder is continuously purged to serve as feedstock for the cement industry, and fresh, highly porous limestone is economically fed back into the system.

I. KIẾN THỨC CƠ BẢN CẦN CÓ (Core Knowledge Required)

Bạn cần chứng minh mình có nền tảng cực kỳ vững chắc về **Kỹ thuật Hóa học (Chemical Engineering)**, **Cơ học Lưu chất (Fluid Mechanics)** và **Nhiệt động lực học (Thermodynamics)**:

1. **Hiện tượng truyền khối và truyền nhiệt (Mass & Heat Transfer):**
    - **Truyền khối (Mass Transfer):** Cách khí CO_{2} khuếch tán từ dòng không khí loãng (420 ppm) qua lớp biên (boundary layer) vào sâu bên trong các lỗ rỗng (pores) của khối $Ca(OH)_2$. Bạn cần hiểu rõ Định luật khuếch tán Fick, Knudsen diffusion (khuếch tán trong lỗ rỗng nano/vi mô).

Fick's Laws of Diffusion

Fick's laws describe normal diffusion. This happens when gas or liquid molecules move through a relatively large space. The molecules collide mostly with each other, not with the walls of the container.

2. **Mechanism**: Driven by a concentration gradient. Molecules move from high concentration to low concentration.
3.

- **Fick's First Law**: Flux is proportional to the concentration gradient.

$$\text{Flux } (J) = -D \cdot \frac{dC}{dx}$$

  - J: Diffusion flux
  - D: Diffusion coefficient
  - dC/dx: Concentration gradient

4.

Where it happens: Large pores, open air, or bulk liquids.

Bạn đã nói: giải thích bằng tiếng anhĐịnh luật khuếch tán Fick, Knudsen diffusion (khuếch tán trong lỗ rỗng nano/vi mô).

Here is the clear explanation of Fick's Laws of Diffusion and Knudsen Diffusion in English.

Fick's Laws of Diffusion

Fick's laws describe normal diffusion. This happens when gas or liquid molecules move through a relatively large space. The molecules collide mostly with each other, not with the walls of the container.

- **Mechanism**: Driven by a concentration gradient. Molecules move from high concentration to low concentration.
- **Fick's First Law**: Flux is proportional to the concentration gradient.
  \(\text{Flux\ }(J)=-D\cdot \frac{dC}{dx}\)
  - \(J\): Diffusion flux
  - \(D\): Diffusion coefficient
  - \(dC/dx\): Concentration gradient
- **Where it happens**: Large pores, open air, or bulk liquids.

Knudsen Diffusion

Knudsen diffusion happens when the size of the pore is extremely small (nano or micro-scale). Specifically, it occurs when the pore diameter is smaller than the mean free path of the gas molecules.

- **Mean Free Path**: The average distance a molecule travels before hitting another molecule.
- **Mechanism**: Molecules collide with the pore walls much more often than with each other. The resistance to movement comes from the walls, not from other molecules.
- **Knudsen Number (\(Kn\))**: Defined as \(Kn = \lambda / d\), where \(\lambda \) is the mean free path and \(d\) is the pore diameter. Knudsen diffusion dominates when \(Kn > 10\).
- **Where it happens**: Nanoporous materials, zeolites, catalysts, and shale rock formations.

Feature	Fick's Diffusion	Knudsen Diffusion
Main Collisions	Molecule-to-Molecule	Molecule-to-Wall
Pore Size	Large (Macropores)	Extremely small (Nanopores)
Pressure Effect	Dependent on pressure	Independent of total pressure
Molecule Weight	Less affected by weight	Highly dependent on molecular weight ($D_K \propto 1/\sqrt{M}$)

   - **Truyền nhiệt (Heat Transfer):** Phản ứng Calcination (nung) cần cung cấp một lượng nhiệt khổng lồ ở ~1,000°C. Phản ứng Hydration lại tỏa nhiệt mạnh. Bạn phải làm chủ các mô hình truyền nhiệt: dẫn nhiệt qua chất rắn, đối lưu nhiệt từ dòng khí nung, và bức xạ nhiệt (radiation) – yếu tố cực kỳ quan trọng trong lò nung nhiệt độ cao.

5. **Động hóa học và Thiết kế Phản ứng (Chemical Kinetics & Reactor Design):**
   - Hiểu sâu về mô hình phản ứng Giữa Chất khí và Chất rắn (Gas-Solid non-catalytic reactions). Hai mô hình kinh điển bạn bắt buộc phải biết là: **Shrinking Core Model** (Mô hình lõi co hẹp) và **Porous Grain Model** (Mô hình hạt xốp).

   - Hiểu cách các yếu tố như nhiệt độ, áp suất, độ ẩm, và diện tích bề mặt (BET surface area) ảnh hưởng đến tốc độ phản ứng (Reaction rate).

6. **Cơ học chất rắn dạng hạt (Particulate Solids & Multiphase Flow):**
   - Vì Rivan di chuyển hạt/khối canxi liên tục, bạn cần kiến thức về: lưu chất hóa (fluidization), dòng chảy của chất rắn hạt (granular flow), hiện tượng tụ áp (pressure drop) qua lớp hạt (định luật Ergun).
7. **Kỹ năng Mô hình hóa (Modeling Skills):**
   - Sử dụng thành thạo các công cụ mô phỏng quy trình như **Aspen Plus/HYSYS** (cho cân bằng vật chất & năng lượng tổng thể), kết hợp lập trình **Python/MATLAB** để tự xây dựng các phương trình vi phân động học riêng cho các thiết bị chưa có sẵn trong thư viện thương mại.

II. CÁC BÀI TOÁN BẠN SẼ PHẢI GIẢI QUYẾT TRONG HỆ THỐNG CỦA RIVAN

Dựa trên triết lý của Rivan (Năng lượng mặt trời off-grid, tối ưu chi phí phần cứng, hệ thống mô-đun hóa trong container), bạn sẽ phải đối mặt với các bài toán cốt lõi sau:

1. Bài toán Carbonation (Hấp thụ CO_{2} từ không khí)

- **Vấn đề cần giải:** Xác định xem điều gì đang kìm hãm tốc độ bắt CO_{2} (Rate-limiting mechanism). Tốc độ chậm là do phản ứng hóa học trên bề mặt canxi chậm (Kinetics)? Do CO_{2} không khuếch tán kịp qua lớp vỏ đá vôi vừa hình thành (Diffusion)? Hay do quạt thổi gió quá nhanh/chậm (Convection)?
- **Nhiệm vụ của bạn:** Xây dựng mô hình dự đoán chính xác sau bao nhiêu giờ thì khối Canxi đạt trạng thái "ngậm no" CO_{2} (giai đoạn bão hòa) để hệ thống biết lúc nào cần đẩy khối đó sang lò nung. Bạn cũng cần tối ưu lượng nước phun ẩm (Humidification) – phun bao nhiêu là đủ để kích hoạt phản ứng mà không làm ướt sũng hệ thống gây tốn năng lượng sấy.

2. Bài toán Calcination (Lò nung tái sinh vôi và giải phóng CO_{2})

- **Vấn đề cần giải:** Đây là thiết bị ngốn năng lượng nhất hệ thống. Rivan chạy bằng điện mặt trời off-grid, nghĩa là nguồn nhiệt có thể biến động theo thời gian trong ngày.
- **Nhiệm vụ của bạn:** Bạn sẽ phải giúp công ty đưa ra quyết định kiến trúc (Drive architecture decisions): Nên dùng **Rotary Kiln** (Lò quay giống lò xi măng - truyền nhiệt chậm nhưng bền, xử lý được hạt lớn) hay **Flash Calciner** (Lò nung nhanh - hạt canxi rơi

tự do, truyền nhiệt siêu nhanh bằng bức xạ nhưng yêu cầu bột phải cực mịn)? Bạn phải lập mô hình để tính toán hiệu suất nhiệt, giảm thiểu thất thoát nhiệt ra vỏ lò.

3. Bài toán Hydration (Trộn vôi sống với nước)

- **Vấn đề cần giải:** Phản ứng $CaO + H_2O \rightarrow Ca(OH)_2$ tỏa nhiệt rất mạnh. Nếu không kiểm soát tốt, nhiệt độ tăng cao sẽ làm biến đổi cấu trúc tinh thể, gây ra hiện tượng thiêu kết (sintering), làm giảm vĩnh viễn độ xốp của vật liệu cho chu kỳ sau.
- **Nhiệm vụ của bạn:** Thiết kế mô hình điều khiển tốc độ cấp nước và giải nhiệt để tạo ra thành phẩm Canxi hydroxit có độ xốp (porosity) tối ưu nhất, đồng thời tận dụng lượng nhiệt tỏa ra này để phục vụ cho các công đoạn khác (Heat integration).

4. Bài toán Nâng cấp quy mô (Scaling laws: Lab $\rightarrow$ Pilot $\rightarrow$ Megawatt)

- **Vấn đề cần giải:** Một mô hình chạy đúng ở quy mô phòng thí nghiệm (vài gam bột) thường sẽ sai hoàn toàn khi lên quy mô Pilot (vài trăm kg) hoặc Megawatt (hàng tấn). Diện tích bề mặt đổi, dòng khí đổi, nhiệt độ phân bố không đều (Thermal gradient).
- **Nhiệm vụ của bạn:** Bạn phải tìm ra các **Hệ số không thứ nguyên (Dimensionless numbers)** như Reynolds, Nusselt, Sherwood để thiết lập định luật tỷ lệ (scaling laws). Giúp đội ngũ kỹ sư cơ khí tự tin chế tạo các bồn chứa kích thước container mà vẫn đảm bảo phân phối khí đều, không có "vùng chết" (dead zones).

💡 LỜI KHUYÊN KHI TRẢ LỜI PHỎNG VẤN (Ghi điểm với Rivan)

- **Nhấn mạnh tư duy "Simple first" (Đơn giản trước tiên):** JD có ghi *"Develop simple, robust models first, and only increase complexity when justified"*. Khi phỏng vấn, hãy thể hiện rằng bạn không lao vào viết các mô hình CFD (Mô phỏng thủy động lực học 3D) phức tạp ngay từ đầu. Hãy bắt đầu bằng bảng tính Excel hoặc script Python 1D (Cân bằng khối lượng/năng lượng cơ bản) để có kết quả nhanh, sau đó mới tăng độ phức tạp khi số liệu thực nghiệm yêu cầu.
- **Tư duy hướng thực nghiệm (Data-driven):** Bạn không ngồi tháp ngà làm toán. Hãy nói về cách bạn sẽ nói chuyện với Đội thực nghiệm (Experimental team) để lấy dữ liệu thô (raw data), điều chỉnh các tham số (model validation) nhằm giúp mô hình ngày càng tiệm cận thực tế.
- **Hiểu về "Sorbent Degradation" (Sự suy giảm vật liệu):** Hãy chủ động nhắc đến việc cấu trúc Canxi sẽ bị chai/giảm hoạt tính sau nhiều chu kỳ nung-hấp thụ và mô hình của bạn sẽ tính toán đến yếu tố suy giảm này để tối ưu hóa chi phí vận hành (OpEx).

If the system utilizes **large macroscopic structured blocks** (such as bricks or honeycomb structures—known as *Structured Sorbents* or *Monoliths*), you **cannot** apply traditional Fluidized Bed or free-flowing Granular Flow technologies. In this scenario, the system must operate using a **Conveyor/Moving Belt mechanism** or a gravity-driven, slow-moving bed framework.

Below is a detailed breakdown of the dimensions, fluid mechanics of these blocks, and how you can apply "first-principles" thinking to solve Rivan's architectural challenges:

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1. Realistic Dimensions of the Calcium Hydroxide ($Ca(OH)_2$) Blocks

In commercial DAC systems utilizing this approach, the calcium structures are never made as thick as standard construction bricks. A brick that is too thick drastically increases the **internal diffusion length**, which isolates the inner core and prevents it from interacting with the ultra-dilute CO_2 in the air.

The optimal dimensions are typically engineered as:

- **Structured Monoliths/Honeycomb Sheets:** Approximately **1 - 3 cm** thick, with length and width matching the scale of the air contactor fan walls. They are perforated with thousands of parallel micro-channels to let air glide through smoothly.
- **Pellets/Briquettes:** Approximately **5 - 20 mm** in diameter (similar to small fuel pellets or animal feed pellets) stacked loosely in layers.

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2. Why You Cannot Fluidize These Large Blocks

As a modeling engineer, you must use **first-principles** to prove to the interviewer why large blocks invalidate fluidization:

- **Minimum Fluidization Velocity (\(U_{mf}\)):** According to the Ergun equation, larger and heavier particles require exponentially higher gas velocities to lift the material into a fluidized, liquid-like state.
- If you attempt to fluidize $Ca(OH)_2$ blocks weighing several hundred grams to kilograms, the pressure drop and required volumetric airflow would be **astronomical**. This directly violates Rivan's core philosophy: *low capital cost, low energy consumption, and off-grid solar operations*.
- **Mechanical Attrition:** The violent, high-velocity collisions inside a fluidized bed would instantly **shatter** the highly porous, brittle calcium structures into fine dust, blowing them straight out of the reactor.

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3. The Architecture Battle: Fluidized Bed vs. Moving Bed (Conveyor)

The JD explicitly states you will *"Drive architecture decisions (e.g. fluidized vs moving bed)"*. This is your opportunity to logically analyze the trade-offs for their specific system:

Option A: Fine Powder + Fluidized Bed System

- **Mechanism:** Calcium is kept as a fine powder or sand-like particles (ranging from tens to hundreds of micrometers).
- **Pros:** Exceptionally fast heat and mass transfer. During calcination, the fine powder absorbs heat and releases CO_2 almost instantaneously.
- **Cons:** Very high pressure drop if the bed is thick. Fine particles are highly prone to elutriation (being blown away), which requires complex, expensive cyclone separators and baghouse filters.

Option B: Porous Blocks + Moving Bed / Conveyor Belt (Rivan's Likely Direction)

- **Mechanism:** Calcium is shaped into porous blocks or large pellets placed on a continuous conveyor belt or fed slowly downward via gravity.
- **Pros:** Air flows freely around the block geometries, resulting in a **pressure drop near zero**. Fans consume minimal electricity, and there is zero material loss from wind entrainment.
- **Cons:** Heat transfer into the solid core is slow. The Calcination kiln must be physically larger or require a much longer residence time to allow the 1,000°C thermal energy to conduct deep into the blocks.

Interview Strategy: How to Answer and Stand Out

When the interviewer asks about making architectural choices or modeling these moving components, structure your response using this logical sequence:

1. **Challenge Geometric Dimensions via First-Principles:** *"Based on mass transfer principles, I wouldn't design the calcium blocks to be too thick, as that introduces severe internal diffusion limitations. I would model the optimal thickness—likely under a few centimeters—to ensure CO_2 can fully penetrate to the core within the allocated residence time."*

2. **Define the Transport Mechanism Clearly:** *"With a structured block or pellet design, we can immediately rule out a traditional fluidized bed due to the unrealistic fluidization velocities and mechanical attrition risks. Instead, I would build a 1D/2D model for a **Moving Bed Reactor** or a conveyor-driven system, where the solid transport velocity is decoupled from the gas velocity."*
3. **Pinpoint the Core Modeling Bottlenecks:** *"Because the solids are large and moving on a belt, the rate-limiting step in the carbonator will be a combination of **External & Internal Mass Transfer** (gas diffusing into the pores). In the calciner, the definitive bottleneck will be **Radiative & Conductive Heat Transfer**—how quickly we can drive 1,000°C of heat into the center of the block to trigger the chemical decomposition."*

This precise, physically grounded approach proves that you understand the real-world mechanics of Rivan's setup, rather than just knowing how to run commercial software theoretically.

To model the **Carbonation Process** (the CO_2 capture step) for Rivan's moving-bed/conveyor system from first principles, we must model a gas-solid reaction where ambient air flows across a moving bed of moist, porous Calcium Hydroxide ($Ca(OH)_2$) blocks.

Unlike the high-temperature calciner where radiation dominates, the carbonator operates at ambient temperature and relies heavily on **external mass transfer (convection)**, **internal pore diffusion**, and **chemical kinetics catalyzed by moisture**.

Here is the first-principles mathematical framework for the Carbonation Process, structured for a technical whiteboard interview.

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1. Physical System & Key Assumptions

- **System Setup:** Porous ($Ca(OH)_2$ blocks of characteristic thickness $2W$ sit on a conveyor belt moving horizontally along the x-axis at velocity v_{s}. Ambient air containing $C_{CO2, \infty} \approx 420 \text{ ppm}$ is blown vertically or cross-flow through the bed.
- **Steady-State:** The system is at a steady state relative to the reactor coordinates. Change over distance (dx) corresponds to change over residence time ($dt = dx/v_s$).
- **Moisture Catalyst:** A controlled water film exists inside the pores, meaning the reaction occurs via the dissolution of CO_{2} into water, reacting with dissolved $Ca(OH)_2$.

---

2. Equation 1: Bulk Gas-Phase Carbon Balance (Air Side)

As ambient air passes through the moving bed along the contactor height (z-axis), CO_2 is continuously stripped out by the calcium blocks. The mass balance for CO_2 in the gas phase is:

$$v_{air} \cdot \frac{dC_{CO2,g}}{dz} = -k_m \cdot a_v \cdot \left(C_{CO2,g} - C_{CO2,s} \right)$$

- v_{air}: Superficial velocity of ambient air blown by the fans $(\mathrm{m/s})$.
- $C_{CO2,g}$: Concentration of CO_2 in the bulk air stream $(\mathrm{mol/m^3})$.
- $C_{CO2,s}$: Concentration of CO_2 at the external surface of the calcium block $(\mathrm{mol/m^3})$.
- k_m: External mass transfer coefficient $(\mathrm{m/s})$, typically derived from the Sherwood number $(Sh = f(Re, Sc))$.
- a_v: Specific surface area of the bed geometry per unit volume $(\mathrm{m^2/m^3})$.

Equation 2: Internal Pore Diffusion & Reaction (Solid Side)

Once CO_2 reaches the surface of the block ($C_{CO2,s}$), it must diffuse inward along the block's internal thickness (y-axis) through the intricate network of micro-pores while simultaneously being consumed by the carbonation reaction.

This is governed by the **Reaction-Diffusion Equation** inside the porous matrix:

$$D_{eff} \cdot \frac{\partial^2 C_{CO2,pore}}{\partial y^2} - r_{rxn} = 0$$

Where:

- $C_{CO2,pore}$: Local concentration of CO_2 inside the pores at depth y.
- D_{eff}: Effective diffusion coefficient (m^2/s), capturing the restriction of the structural pores:

$$D_{eff} = D_{CO2-air} \cdot \frac{\epsilon_p}{\tau}$$

(Here, ϵ_p is the block's porosity and τ is the tortuosity. If Rivan compacts the block too tightly, ϵ_p collapses, driving $D_{eff} \to 0$).

- r_{rxn}: Local rate of consumption of CO_2 by the chemical reaction ($\mathrm{mol/m}^3 \cdot \mathrm{s}$).

Equation 3: Sorbent Conversion & Kinetic Evolution

The local reaction rate (\(r_{rxn}\)) depends heavily on the remaining unreacted \(Ca(OH)_2\) solid fractions. As the block transforms from \(Ca(OH)_2\) into \(CaCO_{3}\), the local solid conversion (\(X_{local}\)) evolves.

Using a first-order approximation with respect to gas concentration and accounting for structural changes:

$$r_{rxn} = k_0 \cdot \exp\left(-\frac{E_a}{RT}\right) \cdot f(\text{Moisture}) \cdot (1 - X_{local}) \cdot C_{CO2,pore}$$

As the conveyor moves the blocks along the reactor length (x-axis), the cross-sectional average conversion of the block ($\bar{X}$) tracks the total carbon captured:

$$\frac{d\bar{X}}{dx} = \frac{1}{v_s \cdot \rho_{solid}} \cdot \left[\frac{1}{W}\int_0^W r_{rxn}(y)\, dy\right]$$

- $\bar{X}$: Average carbonation conversion of the block (0 at the entrance, target $\rightarrow 0.7 - 0.8$ before moving to the calciner).
- $f(\text{Moisture})$: An empirical or thermodynamic scaling factor accounting for the humidification step. Without optimal moisture, this factor drops to near zero.
- ρ_{solid}: Molar density of the calcium sorbent (mol/m^3).

1. **Define the Core Boundary Condition Interlock:** *"The external mass transfer equation (Eq. 1) and internal pore diffusion equation (Eq. 2) couple perfectly at the block's boundary surface ($y = W$), where the mass flux leaving the air must equal the mass flux entering the solid: $-D_{eff}\frac{\partial C_{CO2}}{\partial y}\Big|_{y=W} = k_m(C_{CO2,g} - C_{CO2,s})$."*

2. **Expose the Pore-Plugging Phenomenon via Thiele Modulus:** *"To address Rivan's concern about low porosity due to over-compaction, I will track the* ***Thiele Modulus (ϕ)*** *of the system: $\phi = W\sqrt{\frac{k}{D_{eff}}}$. If we over-compact the blocks, D_{eff} decreases sharply, driving $\phi \gg 1$. This mathematically indicates a severe internal diffusion limitation, meaning CO_2 will instantly react at the crust ($y = W$) and choke off the core, dropping our overall conversion capacity ($\bar{X}$)."*

3. **Drive Architecture Decisions (Simplifying First):** *"In line with Rivan's philosophy of developing simple models first, I will initially solve this system by assuming an isothermal 1D moving slice. We can use this to optimize the fan velocity (v_{air}) against the belt speed (v_s) to minimize the energy consumed per kilogram of CO_2 captured."*

Here is the first-principles mathematical framework for the **Calcination Process** (the high-temperature regeneration step) for Rivan's conveyor/moving-bed system, completely written in English and tailored for your whiteboard interview.

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1. Physical System & Key Assumptions

- **System Setup:** The carbonated blocks (now primarily composed of limestone, $CaCO_3$) travel on a moving belt horizontally along the x-axis at velocity v_{s} through a kiln of length L.
- **Thermal Environment:** The kiln is sealed to maintain a pure atmosphere of released CO_2. It is heated to $T_{wall} \approx 1000^\circ\text{C}$ using clean, off-grid electrical resistance or radiative elements.
- **Dominant Mechanisms:** At this high temperature, **radiative heat transfer** to the surface of the block and **thermal conduction** through the solid matrix are the dominant rate-limiting mechanisms.

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2. Equation 1: Solid-Phase Energy Balance (Thermal Control)

The thermal energy absorbed by the moving solid blocks must supply the sensible heat needed to raise the blocks' temperature and provide the massive heat of reaction ($\Delta H_{rxn} \approx 178 \text{ kJ/mol}$) required for this endothermic decomposition.

For a differential length of the kiln dx, the energy balance of the solid stream is:

$$\dot{m}_s \cdot C_{p,s} \cdot \frac{dT_s}{dx} = q_{rad} + q_{conv} - \Delta H_{rxn} \cdot \left(- \frac{d\dot{m}_{CaCO3}}{dx} \right)$$

- $\dot{m}_s$: Total mass flow rate of the solid blocks traveling on the belt (kg/s).
- $C_{p,s}$: Effective specific heat capacity of the solid matrix (J/kg·K), which shifts as $CaCO_3$ converts to CaO.
- T_s: Cross-sectional average temperature of the solid block at position x (K).
- $-\frac{d\dot{m}_{CaCO3}}{dx}$: The mass consumption rate of $CaCO_3$ due to chemical decomposition along the length of the kiln (kg/m·s).

Heat Input Terms (q_{rad} and q_{conv}):
Because the kiln operates at nearly 1000°C, radiative heat transfer vastly dominates over convection:

$$q_{rad} = \epsilon \cdot \sigma \cdot A_s \cdot \left(T_{wall}^4 - T_{s,surf}^4\right)$$

- ε: Emissivity of the calcium-based block; σ: Stefan-Boltzmann constant; A_s: Exposed surface area of the bed per unit length; $T_{s,surf}$: External surface temperature of the block.

Equation 2: Shrinking Core Kinetics (Chemical Decomposition)

Because the blocks are macroscopic structures (centimeter scale), they do not decompose uniformly. The high thermal flux creates an outer shell of regenerated lime (CaO), while the unreacted limestone ($CaCO_3$) core shrinks inward over time.

The localized reaction rate of the block along the kiln axis x is governed by Arrhenius kinetics and is heavily suppressed if the local partial pressure of CO_2 matches the thermodynamic equilibrium pressure (\(P_{eq}\)):

$$\frac{dX_{calc}}{dx} = \frac{k_0}{v_s} \cdot \exp\left(-\frac{E_a}{RT_s}\right) \cdot \left(1 - \frac{P_{CO2,pore}}{P_{eq}(T_s)}\right) \cdot (1 - X_{calc})^{2/3}$$

- X_{calc}: Calcination conversion fraction (0 at the kiln entrance, target → 1.0 at the exit).
- $(1 - X_{calc})^{2/3}$: A geometric term representing the shrinking surface area of the unreacted $CaCO_3$ core interface.
- $P_{eq}(T_s)$: The equilibrium pressure of CO_2 over $CaCO_3$, which increases exponentially with temperature: $P_{eq} = \exp\left(A - \frac{B}{T_s}\right)$. If T_s is too low or $P_{CO2,pore}$ is too high, the term $(1 - P_{CO2}/P_{eq})$ drops to zero, completely stalling the reaction.

Equation 3: Internal Heat Conduction & Gas Escape (The "Large Block" Barrier)

Inside the moving block, heat must conduct inward along its thickness (y-axis), while the generated pure CO_2 gas must diffuse outward through the newly formed CaO porous crust:

- **Thermal Gradient (Fourier's Law):**

$$\rho_s \cdot C_{p,s} \cdot v_s \cdot \frac{\partial T_s}{\partial x} = \lambda_{eff} \cdot \frac{\partial^2 T_s}{\partial y^2} + r_{rxn} \cdot (-\Delta H_{rxn})$$

(Where λ_{eff} is the effective thermal conductivity of the porous calcium matrix. If the blocks are designed with poor thermal conductivity, the center remains cold and unreacted).

- **Gas Outflow (Darcy's Law / Fickian Diffusion):**
  Because CO_2 gas is generated rapidly inside the core, a pressure gradient builds up to push the gas out through the pores:

$$N_{CO2} = -\frac{K_{permeability}}{\mu} \cdot \frac{\partial P_{CO2}}{\partial y}$$

Interview Delivery Strategy: How to Explaining this Calciner Model

When standing at the whiteboard presenting this calciner model, use this professional engineering narrative to showcase your skills:

1. **Highlight the Interlocking Bottlenecks (Heat vs. Mass Transfer):** *"The calciner model exposes a fascinating coupled physical bottleneck. The reaction requires a massive heat influx (Eq. 1), but as the outer crust turns into CaO, it acts as a thermal insulator (low \(\lambda _{eff}\)), slowing down heat conduction to the core. Concurrently, if the CO_2 cannot escape the pores quickly enough (Eq. 3), \(P_{CO2,pore}\) will spike, approaching \(P_{eq}\) and thermodynamically shutting down the reaction kinetics (Eq. 2)."*
2. **Demonstrate Architecture Decision Capabilities:** *"This model is exactly how I would drive the choice between a rotary kiln, a flash calciner, or Rivan's moving-bed approach. If our model shows that internal heat conduction across a 2-cm block takes too long—requiring an excessively long and expensive kiln—I will advise the design team to introduce internal hollow micro-channels (honeycomb structures) into the machine-*

extruded blocks. This shortens both the thermal conduction pathway and the gas diffusion distance without changing the external dimensions of the moving bed."

3. **Align with Rivan's "Simple First" Philosophy:** *"To execute this quickly, I would first solve this as an analytical 1D lumped-parameter model along the kiln length x. This allows us to map the initial operating envelope, pairing the electrical solar power input against the conveyor velocity v_{s} before we invest time in building computationally expensive 3D finite-element models."*

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2. AN ALKALINE WATER ELECTROLYSER (H_{2})

An alkaline electrolyser splits water into Hydrogen (H_{2}) and Oxygen (O_{2}) using a liquid electrolyte (typically 20-30% KOH) and electricity from Rivan's off-grid solar arrays.

A. Physical System & Key Assumptions

- **System Setup:** A multi-cell stack where water is fed to the cathode side. Hydroxide ions (OH^{-}) pass through a porous diaphragm separator.
- **Core Phenomenon:** The process is governed by **electrochemical kinetics (Butler-Volmer equation)**, **Ohmic resistance (voltage losses)**, and **multiphase fluid dynamics** (gas bubbles blocking the electric current).

B. Equation 1: Cell Voltage Efficiency & Energy Balance (The Voltage Stack)

The total operating voltage of a single cell (V_{cell}) is higher than the ideal thermodynamic voltage due to internal losses (overpotentials). Modeling this determines the energy efficiency of the H_{2} production:

$$V_{cell} = V_{rev}(T, P) + \eta_{act,an} + \eta_{act,cat} + \eta_{ohmic}$$

Where:

- $V_{rev}(T, P)$: Reversible cell voltage (≈ 1.23 V under standard conditions, but decreases as operating temperature T increases).
- η_{act}: Activation overpotential at the anode (an) and cathode (cat), representing the energy barrier to kickstart the chemical reaction. It is modeled using the **Butler-Volmer / Tafel equation**:

$$\eta_{act} = \frac{RT}{\alpha F} \ln\left(\frac{j}{j_0}\right)$$

(where j is the current density A/m^2, and j_0 is the exchange current density).

- η_{ohmic}: Ohmic overpotential due to resistance in the KOH electrolyte, electrodes, and diaphragm:

$$\eta_{ohmic} = j \cdot R_{internal} = j \cdot \left(\frac{\delta_{electrolyte}}{\sigma_{electrolyte}}\right)$$

C. Equation 2: Mass Balance & Hydrogen Production Rate (Faraday's Law)

The total molar flow rate of Hydrogen produced ($\dot{n}_{H2}$, in mol/s) is directly proportional to the total electrical current supplied (I, in Amperes) and the Faraday efficiency (η_F):

$$\dot{n}_{H2} = \eta_F \cdot \frac{I \cdot N_{cells}}{2F}$$

Where:

- N_{cells}: Number of individual cells stacked in series.
- F: Faraday's constant ($96,485$ C/mol).

Where:

- N_{cells}: Number of individual cells stacked in series.
- F: Faraday's constant ($96,485$ C/mol).
- 2: Number of electrons transferred per molecule of H_2.
- η_F: Faraday efficiency (≤ 1.0), which captures gas parasitic losses (e.g., hydrogen gas dissolving back into the liquid electrolyte and crossing over the membrane).

"Since Rivan uses intermittent, off-grid solar energy, the current density (\(j\)) will fluctuate throughout the day. My model must capture how fluctuating power affects the cell temperature

(\(T\)) and Faraday efficiency (\(\eta _{F}\)). At low solar power (low \(j\)), gas crossover becomes dominant, causing \(\eta _{F}\) to drop significantly and creating safety risks. I will model the thermal mass of the system to optimize the startup time and dynamic load-following capability of the stack."

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3. A SABATIER REACTOR (CH_4)

The Sabatier reactor combines the captured CO_2 and the generated H_2 over a catalyst (typically Nickel or Ruthenium) to synthesize Methane (CH_4) and Water (H_2O).

A. Physical System & Key Assumptions

- **System Setup:** A fixed-bed catalytic reactor packed with solid catalyst pellets. Reactant gases ($CO_2 + 4H_2$) flow through the bed.
- **Core Phenomenon:** The reaction ($CO_2 + 4H_2 \rightleftharpoons CH_4 + 2H_2O$) is **highly exothermic** ($\Delta H_{rxn} \approx -165\ \mathrm{kJ/mol}$). If heat is not removed instantly, the catalyst will overheat (thermal runaway) or deactivate.

B. Equation 1: Reaction Kinetics & Conversion Rate

The rate of CO_2 consumption ($r_{Sabatier}$, in $\mathrm{mol/m^3 \cdot s}$) is modeled using Langmuir-Hinshelwood-Hougen-Watson (LHHW) kinetics, which accounts for gas molecules adsorbing onto the solid catalyst surface:

$$r_{Sabatier} = \frac{k(T) \cdot \left(P_{CO2} P_{H2}^4 - \frac{P_{CH4} P_{H2O}^2}{K_{eq}(T)} \right)}{\left(1 + K_{CO2} P_{CO2} + K_{H2} P_{H2} + K_{H2O} P_{H2O}\right)^5}$$

Where:

- $k(T)$: Arrhenius rate constant ($k_0 \exp(-E_a/RT)$).
- $K_{eq}(T)$: Thermodynamic equilibrium constant. **Crucial point:** Because the reaction is exothermic, K_{eq} *decreases* as temperature rises. If the reactor gets too hot, thermodynamics will prevent high methane conversion.

C. Equation 2: Reactor Heat Balance (The Temperature Control Bottleneck)

To prevent catalyst damage, we must model the 1D/2D temperature profile along the reactor length (x):

$$\rho_g \cdot C_{p,g} \cdot v_g \cdot \frac{dT_g}{dx} = \rho_{bed} \cdot r_{Sabatier} \cdot (-\Delta H_{rxn}) - \frac{4U}{D_{tube}} \cdot \left(T_g - T_{coolant}\right)$$

Where:

- $\rho_g, C_{p,g}, v_g$: Density, heat capacity, and velocity of the gas mixture.
- ρ_{bed}: Bulk density of the catalyst bed.
- U: Overall heat transfer coefficient to the cooling jacket.
- D_{tube}: Diameter of the reactor tube.

The Sabatier reactor presents a classic chemical engineering trade-off: Kinetics vs. Thermodynamics. High temperature gives us fast reaction rates (Eq. 2), but thermodynamic equilibrium (Eq. 1) severely limits methane purity at high temperatures, leaving unreacted CO_{2} and H_{2} in the output. Since Rivan aims for a 99.9% pure Synthetic Natural Gas (SNG) to inject straight into the grid, I would drive the architecture decision toward a multi-stage reactor system with inter-stage cooling or a recirculation loop. The first stage operates hot for high throughput, and the final stages run cooler to shift the thermodynamic equilibrium toward 100% methane."

Since the geometric size of the reactor is fixed, when the solar power drops to 200 kW, I cannot change the physical dimensions of the kiln. Instead, I will utilize the model to dynamically adjust the **operating envelopes**. Specifically, the model will signal the system to reduce the conveyor belt velocity (v_{s}) by a factor of 5.

As the conveyor slows down, the **residence time** of the material inside the kiln increases proportionally. This ensures that the calcium blocks still receive the total cumulative heat flux required for full decomposition, even though the instantaneous thermal power input at that moment is lower."

Integrating all three highly complex systems (Calcium-Looping DAC, Alkaline Water Electrolyser, and Sabatier Reactor) into a **single, standard shipping container** is a breakthrough for modular logistics, but it introduces massive technical and physical bottlenecks from a first-principles perspective [Rivan].

Here are the core problems they face, which you can highlight during your interview:

1. Extreme Thermal Gradients and Heat Integration

This is the most significant thermodynamic challenge when placing these three distinct processes inside a tight, enclosed space:

- **The Temperature Disconnect:**
    - The Calciner kiln (DAC) operates at an extreme high-temperature regime of **~1,000°C** [Rivan].
    - The Alkaline Electrolyser runs at a low-temperature regime of **~60°C – 90°C**.
    - The Sabatier Reactor operates at a mid-temperature regime of **~250°C – 400°C**.
- **The Problem:** The container will experience an intense localized thermal gradient. Without flawless thermal insulation, heat leaking from the 1,000°C kiln will boil or degrade the KOH electrolyte solution in the adjacent electrolyser, or cause thermal runaway in the Sabatier reactor (which kills the methane conversion equilibrium).
- **The Modeling Opportunity:** This is where your role shines. You must build a **Heat Integration Model** to capture the exothermic waste heat from the

Sabatier reactor and the heat released during the lime-slaking (Hydration) stage, using it to pre-heat the incoming limestone blocks before they enter the calciner, thereby saving solar power.

2. Safety and Hazardous Gas Zoning (Explosion Risks)

Packing these specific technologies into a single metal box severely concentrates safety risks:

- **Dangerous Gas Coexistence:** The electrolyser generates high-pressure **Hydrogen (H_2)**, and the Sabatier reactor synthesizes **Methane (CH_4)**—both are highly flammable and explosive gases. Right next to them, the calciner uses high-voltage, high-temperature electrical heating elements at 1,000°C, which act as a permanent ignition source [Rivan].
- **The Problem:** Even a microscopic structural crack on the H_2 or CH_4 piping caused by transport or thermal stress could leak gas into the high-temperature zone, triggering a catastrophic explosion that destroys the entire container. The system requires strict localized pressure management, leak-detection models, and physical blast-wall zoning within the container.

3. Spatial and Footprint Constraints vs. Mass Transfer Dynamics

- **Volumetric Disproportion:** Ambient Air Capture (DAC) inherently requires processing millions of cubic meters of air because atmospheric CO_2 is ultra-dilute (~420 ppm). Therefore, the air contactor (fans and carbonation beds) must physically have a massive surface area.
- **The Problem:** When you squeeze huge air contactors, moving conveyor systems, the liquid-heavy electrolyser, water treatment units, and the Sabatier reactor into one container, internal space is choked. This creates severe geometric constraints, resulting in high air pressure drops (which overloads the fans) and makes mechanical routing (piping and wiring) incredibly complex to model and maintain.

4. Mechanical Vibrations and Structural Fatigue

- The moving-bed conveyor system, the low-pressure extrusion mechanisms, and the fans are heavy, continuously moving mechanical components that generate constant **vibrations and structural stress** [Rivan].
- Conversely, the alkaline electrolyser stack consists of hundreds of thin, finely compressed electrodes and delicate polymer diaphragms. It is extremely sensitive to mechanical vibration, which can easily wear out the sealing gaskets, leading to dangerous leaks of the highly corrosive liquid KOH electrolyte. Putting a heavy vibrating machine next to a highly sensitive, sealed stack within a single container frame will rapidly accelerate hardware fatigue.

💡 Summary Interview Delivery (Your Verbal Answer)

When asked about the container integration during the interview, you can summarize your thoughts like this:

"Putting all three operations into a single container introduces several critical **first-principles engineering bottlenecks**:

1. **Thermal Management & Extreme Gradients**: We are placing a **1,000°C Calciner** directly next to an **80°C Alkaline Electrolyser** [Rivan]. This requires advanced thermal insulation and strict heat integration modeling to utilize waste heat without cooking the electrolyte.
2. **Safety & Hazard Mitigation**: We have high-pressure Hydrogen (H_2) and Methane (CH_4) operating inside the same enclosed space as a high-temperature ignition source (the kiln) [Rivan]. Managing gas crossover, leak detection, and zoning boundaries within the container is a massive safety challenge.
3. **Footprint vs. Mass Transfer**: DAC requires a massive air-contacting surface area because ambient CO_2 is ultra-dilute. Fitting large air contactors alongside heavy moving-bed mechanics, water treatment units, and the

Sabatier reactor forces severe spatial and volumetric constraints, which can lead to high pressure drops.

4. **Mechanical Vibrations**: The moving-bed conveyor system creates constant mechanical stress and vibrations [Rivan]. If not isolated properly, these vibrations will compromise the structural integrity and sealing gaskets of the electrolyser stack, leading to hazardous KOH leaks."

"To determine how long a $Ca(OH)_2$ or CaO batch lasts, we must model the **Sorbent Degradation Kinetics** across hundreds of cycles. The lifespan is fundamentally limited by three phenomena: **thermal sintering** at 1,000°C which collapses the micro-pores, **irreversible chemical poisoning** from atmospheric SOx forming $CaSO_4$, and **mechanical attrition** from the conveyor movement.

In my model, I will implement a semi-empirical capacity decay equation: $a_N = a_r + (1 - a_r)\exp(-k_{deact} \cdot N)$, where the deactivation constant (k_{deact}) is validated using experimental lab data.

Because Rivan uses cheap limestone, the solution isn't to stop the system to change the bed. Instead, my model will optimize a continuous **Purge and Makeup rate**. By continuously purging a small fraction of degraded fines and making up with fresh, porous limestone, we can maintain a steady-state average sorbent activity inside the container while minimizing the Total Cost of Carbon Capture."

To accurately determine **how long a calcium batch ($Ca(OH)_2$ or CaO) lasts** before it needs to be replaced or replenished inside Rivan's DAC system, you cannot rely on guesswork. Instead, you must build and deploy a **Sorbent Degradation Model**.

During the interview, the panel will be highly impressed if you approach this question by splitting it into two distinct parts: the **Physical Mechanisms (First-principles)** and the **Mathematical Modeling Strategy**.

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1. Core Mechanisms of Sorbent Degradation (First-Principles)

After every single cycle (Absorption → Calcination → Slaking/Hydration), the structural and chemical integrity of the calcium material degrades due to three core physical phenomena:

1. **Thermal Sintering:**
    - At the extreme temperature of the calciner kiln (~1,000°C), the calcium crystals undergo partial surface melting and structural deformation. This causes the internal microscopic pores (micro-pores) to collapse and fuse together.
    - **The Consequence:** The specific surface area (BET surface area) drops sharply after each heating cycle, drastically reducing the material's capacity to trap CO_2.
2. **Chemical Poisoning / Irreversible Deactivation:**
    - Ambient air does not contain just CO_2; it also contains trace impurities like sulfur oxides (SO_{x}) and particulate matter.
    - Calcium reacts with SO_{x} to form calcium sulfate ($CaSO_4$, or gypsum). This reaction is **completely irreversible** at 1,000°C. Over time, a fraction of your active calcium is converted into "dead" gypsum, rendering it useless for carbon capture.
3. **Mechanical Attrition and Structural Fragmentation:**
    - Due to continuous mechanical transport along the conveyor belt, combined with the cyclic volumetric expansion and contraction during phase changes ($Ca(OH)_2 \leftrightarrow CaCO_3$), the blocks develop micro-cracks and eventually disintegrate into fine dust (fines).

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2. Mathematical Modeling Approach for Lifespan Prediction

As a process simulation engineer, you will program a capacity decay equation based on the total number of operational loops/cycles (N):

A. The Semi-Empirical Sorbent Capacity Decay Equation

The maximum CO_2 capture capacity of the sorbent at the N-th cycle (\(a_{N}\)) is mathematically modeled as:

$$a_N = a_r + (1 - a_r) \cdot \exp(-k_{deact} \cdot N)$$

- a_N: Residual fractional activity at cycle N (e.g., a value of 0.4 means the block can only capture 40% of the CO_2 it could absorb on day one).
- a_r: Residual ultimate capacity—the minimum baseline activity that the material cannot drop below (typically ranges between 0.05 and 0.15).
- k_{deact}: Deactivation rate constant, which is a function of the kiln temperature and impurity concentrations in the air. This parameter is directly validated using experimental lab data.

B. Operational Strategy: Continuous Purge & Makeup Flowrates

Because Rivan runs a continuous looping system, it makes no sense to wait for all the calcium blocks to completely die and then shut down the plant for a full bed replacement. Instead, the operational solution is a **continuous Purge and Makeup strategy**.

Your model will be responsible for calculating the optimal mass flow rate of fresh, cheap limestone input (\(\.{m}_{makeup}\)) and the purging of degraded fine powder (\(\.{m}_{purge}\)) to maintain a financially viable steady-state average activity (\(a_{avg}\)) inside the container (typically targeting an average activity of 30-40%):

$$\dot{m}_{makeup} = \dot{m}_{circulating} \cdot \left(\frac{1 - a_{avg}}{a_{avg} - a_r} \right) \cdot f(k_{deact})$$

To determine how long a $Ca(OH)_2$ or CaO batch lasts, we must model the **Sorbent Degradation Kinetics** across hundreds of cycles. The lifespan is fundamentally limited by three phenomena: **thermal sintering** at 1,000°C which collapses the micro-pores, **irreversible chemical poisoning** from atmospheric SOx forming $CaSO_4$, and **mechanical attrition** from the conveyor movement.

In my model, I will implement a semi-empirical capacity decay equation: $a_N = a_r + (1 - a_r)\exp(-k_{deact} \cdot N)$, where the deactivation constant (k_{deact}) is validated using experimental lab data.

Because Rivan uses cheap limestone, the solution isn't to stop the system to change the bed. Instead, my model will optimize a continuous **Purge and Makeup rate**. By continuously purging a small fraction of degraded fines and making up with fresh, porous limestone, we can maintain a steady-state average sorbent activity inside the container while minimizing the Total Cost of Carbon Capture."

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1. Step 1: Data Gathering from the Lab (Xác định dữ liệu đầu vào từ phòng Lab)

First, I will collaborate closely with the experimental team to extract raw dataset profiles from laboratory-scale equipment. For sorbent degradation, we primarily need data from a **TGA (Thermogravimetric Analyzer)** or a small fluidized bed reactor running automated cyclic carbonation-calcination loops.

The key raw metrics we will extract are:

- **Mass change vs. time/cycle count ($m_N(t)$):** Tracks how much CO_2 is absorbed and released in each cycle N.
- **BET Surface Area & Pore Volume:** Tracks the physical collapse of the microstructure due to sintering.
- **XRD (X-ray Diffraction) mineralogy:** Measures the exact percentage of inactive components like $CaSO_4$ (poisoning).

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2. Step 2: Parameter Calibration & Optimization (Hiệu chuẩn tham số mô hình)

Once the experimental data is collected, I will use regression and optimization algorithms (such as Levenberg-Marquardt or Genetic Algorithms written in **Python/MATLAB**) to fit the experimental curve into our first-principles equations.

- **Fitting the Decay Curve:** We plot the experimental fractional capacity (a_{exp}) against the cycle number (N).
- By running a non-linear least-squares optimization, the script will isolate the exact mathematical values for the **Deactivation Constant (k_{deact})** and the **Residual Capacity (a_{r})** specific to Rivan's unique calcium extrusion formulation:

$$\min_{k_{deact}, a_r} \sum_{N=1}^{N_{max}} \left[a_{exp,N} - \left(a_r + (1 - a_r) e^{-k_{deact} \cdot N} \right) \right]^2$$

3. Step 3: Model Validation & Accounting for Regime Shifts (Chứng thực và xử lý sự khác biệt quy mô)

A crucial challenge is that **lab data is often collected under perfect conditions** (pure CO_2 streams, precise temperature controls, zero mechanical friction). The real pilot system inside the container will experience **regime shifts** (e.g., uneven wind distribution, fluctuating solar heat flux, actual mechanical vibrations).

To translate lab data into a validated model that predicts real system behavior, I will introduce **Scaling and Correction Factors**:

- **Mass Transfer Scaling:** In the lab TGA, there is zero external mass transfer resistance. In the real scale, I will multiply the kinetic rate by an **Effectiveness Factor ($\eta _{eff}$)** derived from the Thiele Modulus (ϕ) to account for gas diffusion limitations within our thick calcium blocks.
- **Attrition Scaling:** I will include a mechanical wear coefficient ($k_{attrition}$) calibrated from the pilot plant's physical filters to account

for the accelerated material loss caused by the moving conveyor system, which lab TGAs cannot measure.

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4. Step 4: Closing the Loop via Discrepancy Resolution (Sửa đổi mô hình liên tục)

As noted in the JD (*"Work closely with experimental and design teams to resolve discrepancies"*), validation is an iterative process.

If the pilot plant captures less CO_2 than my model predicts, I will isolate the root cause by cross-checking individual variables:

- Is the discrepancy caused by chemistry? (Check if local SO_{x} poisoning is higher than expected).
- Is it physics/geometry? (Check if the conveyor is moving too fast or if air is bypassing the blocks due to poor packing/channeling).

I will then refine the model's boundary conditions, update the software code, and re-deploy the updated operational envelope to the design team.

💡 Verbal Interview Answer (Your Pitch to the Panel)

When the interviewer asks: **"How do you translate our lab data into a validated model for the container system?"**, you can deliver this powerhouse response:

"Translating lab data into real-world predictions requires a structured, data-driven pipeline. First, I will take cyclic data from the lab's Thermogravimetric Analyzer (TGA) and run a non-linear optimization in Python to calibrate our core parameters, such as the deactivation rate (k_{deact}) and residual activity (a_{r}).

However, lab data represents an idealized regime. To predict true behavior inside the container, I will scale the lab-derived kinetics using first-principles dimensionless numbers—like the Thiele Modulus—to account for internal diffusion limits within our extruded blocks, and introduce an attrition coefficient for conveyor wear.

Finally, validation is a continuous feedback loop. When discrepancies occur between the pilot plant data and the model, I won't just tweak random fitting parameters. I will systematically audit the model's heat and mass transfer boundary conditions to isolate whether the root cause is chemical deactivation or fluid-dynamics channeling, ensuring our model remains mathematically robust and physically grounded."
